

CODE:

Practical_6.c

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>

#define CLIENT_FIFO "client_to_server"
#define SERVER_FIFO "server_to_client"

int main()
{
    char message[200];
    char response[300];

    /* Create named pipes */
    mkfifo(CLIENT_FIFO, 0666);
    mkfifo(SERVER_FIFO, 0666);

    printf("Server started...\n");
    printf("Waiting for client messages...\n");

    while (1)
    {
        /* Open client FIFO for reading */
        int read_fd = open(CLIENT_FIFO, O_RDONLY);

        if (read_fd == -1)
        {
            perror("open client FIFO");
            exit(EXIT_FAILURE);
        }

        /* Read message from client */
        ssize_t bytes = read(read_fd, message, sizeof(message) - 1);
        close(read_fd);

        if (bytes > 0)
        {
            message[bytes] = '\0';

            printf("Client message: %s\n", message);

            /* Process the message */
            snprintf(response, sizeof(response),
                     "Server received: %s", message);

            /* Open server FIFO for writing */
            int write_fd = open(SERVER_FIFO, O_WRONLY);

            if (write_fd == -1)
```

```

    {
        perror("open server FIFO");
        exit(EXIT_FAILURE);
    }

    /* Send response */
    write(write_fd, response, strlen(response) + 1);

    close(write_fd);

    printf("Response sent to client.\n\n");
}
}

/* Remove FIFOs */
unlink(CLIENT_FIFO);
unlink(SERVER_FIFO);

return 0;
}

```

Practical_6_client.c

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>

#define CLIENT_FIFO "client_to_server"
#define SERVER_FIFO "server_to_client"

int main(int argc, char *argv[])
{
    char message[200];
    char response[300];

    /* Get message from command line */
    if (argc > 1)
    {
        snprintf(message, sizeof(message), "%s", argv[1]);
    }
    else
    {
        printf("Enter message: ");
        fgets(message, sizeof(message), stdin);

        message[strcspn(message, "\n")] = '\0';
    }

    /* Open client FIFO for writing */
    int write_fd = open(CLIENT_FIFO, O_WRONLY);

    if (write_fd == -1)
    {
        perror("open client FIFO");
        exit(EXIT_FAILURE);
    }

```

```

/* Send message */
write(write_fd, message, strlen(message) + 1);

close(write_fd);

printf("Message sent: %s\n", message);

/* Open server FIFO for reading */
int read_fd = open(SERVER_FIFO, O_RDONLY);

if (read_fd == -1)
{
    perror("open server FIFO");
    exit(EXIT_FAILURE);
}

/* Receive server response */
ssize_t bytes = read(read_fd, response, sizeof(response) - 1);

if (bytes > 0)
{
    response[bytes] = '\0';
    printf("Server response: %s\n", response);
}

close(read_fd);

return 0;
}

```

Screen Shots:

```

sri_charan@SRICHARAN: ~/Pr
+ v
sri_charan@SRICHARAN:~$ ls
OSLab OSLab2 Practical 'Practical 1' Skill hello.c
sri_charan@SRICHARAN:~$ cd Practical
sri_charan@SRICHARAN:~/Practical$ gcc Practical6_client.c -o Practical6_client
sri_charan@SRICHARAN:~/Practical$ ./Practical6_client
Enter message: Hello this Sri Charan
Message sent: Hello this Sri Charan
Server response: Server received: Hello this Sri Charan
sri_charan@SRICHARAN:~/Practical$

```

```

sri_charan@SRICHARAN: ~/Pr
+ v
sri_charan@SRICHARAN:~/Practical$ gcc Practical6.c -o Practical6
sri_charan@SRICHARAN:~/Practical$ ./Practical6
Server started...
Waiting for client messages...
Client message: Hello this Sri Charan
Response sent to client.

^C
sri_charan@SRICHARAN:~/Practical$

```